AI in Business Valuation

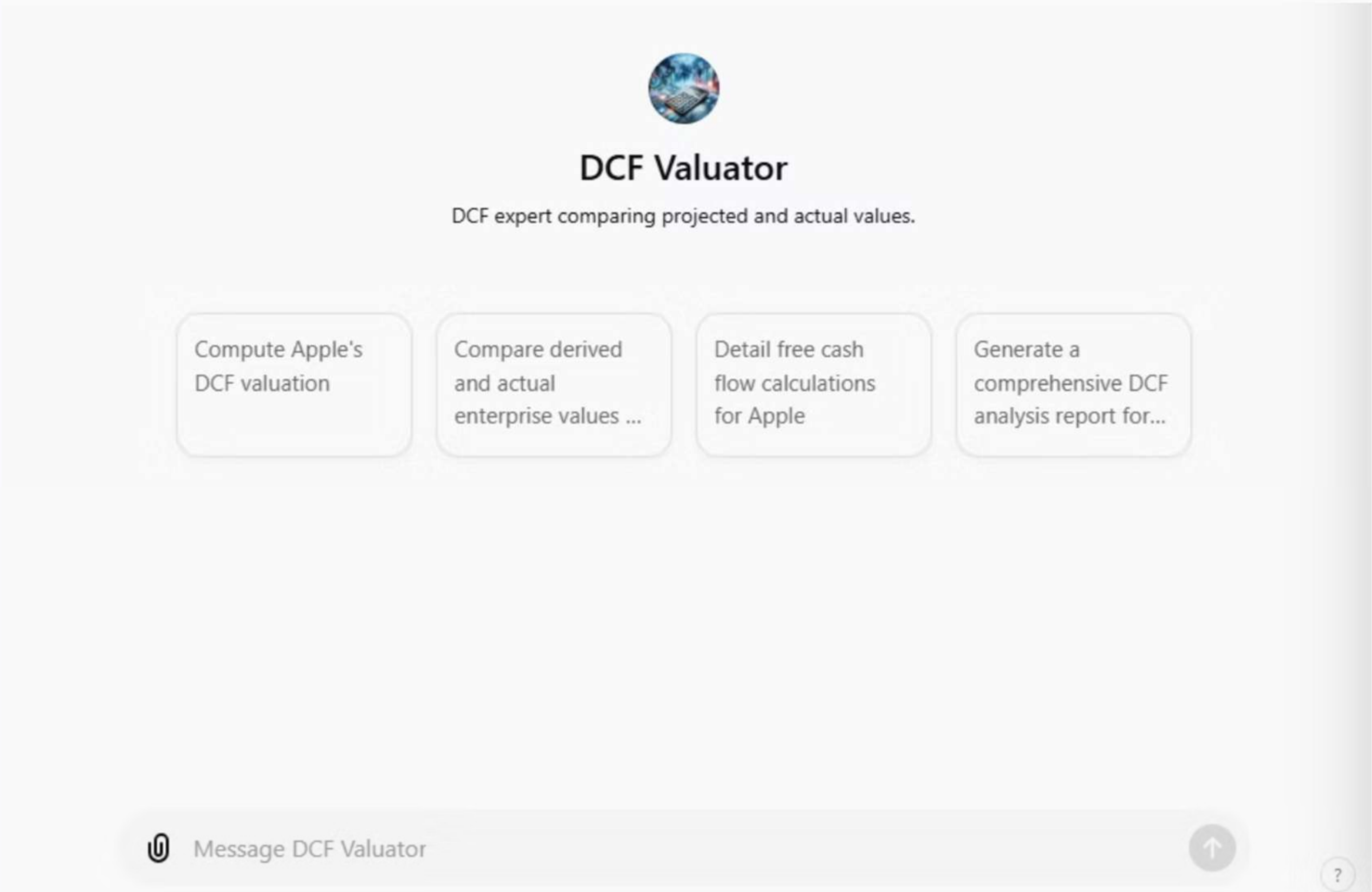
Business Analysis and Valuation

How AI can help: What changes, what doesn't

- Valuation theory: unchanged
  - DCF, multiples, cost of capital, terminal value
- What changes: how inputs get built, checked

How AI Has Evolved: From Prompts to Agents

- 2023: prompt-based AI — a human asks, the model answers
  - Single-turn chat responses; every next step needs a fresh instruction from a person
- Today: agentic AI — the model plans and acts
  - Breaks a goal into steps, calls tools and data sources, checks its own output, and keeps going without a prompt for every step
- Why it matters for valuation work
  - Multi-step tasks — sourcing peer data, building comps, updating models — can now run with far less manual back-and-forth



Build a Custom GPT in OpenAI

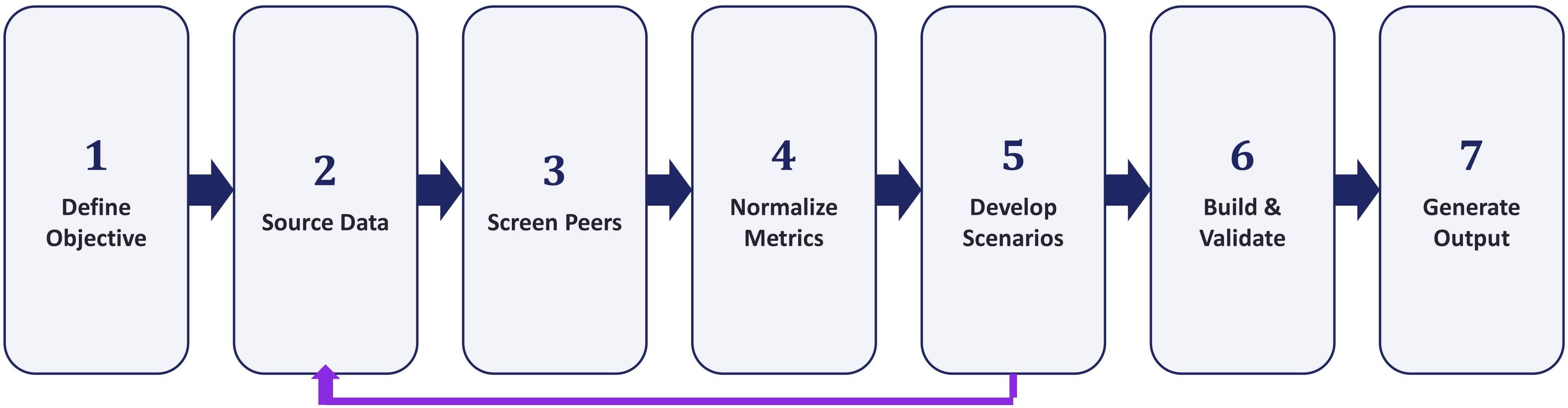
Mistakes made

```python
# Extract relevant figures
net_income = income_stmt.loc['Net Income'][0]
depreciation_amortization = cash_flow_stmt.loc['Depreciation & Amortizat
capex = cash_flow_stmt.loc['Capital Expenditures'][0]

# Calculate Free Cash Flow
free_cash_flow = net_income + depreciation_amortization - capex

# Present in tabular format
data = {'Item': ['Net Income', 'Depreciation & Amortization', 'Capital E
        'Value': [net_income, depreciation_amortization, capex, free_cas

df = pd.DataFrame(data)
print(df)
```

```python
# Extract relevant figures (using most recent annual data)
net_income = income_stmt.loc['Net Income'][0]
depreciation_amortization = cash_flow_stmt.loc['Depreciation & Amortizat
capex = cash_flow_stmt.loc['Capital Expenditures'][0]
change_in_working_capital = cash_flow_stmt.loc['Change In Working Capit

# Calculate Free Cash Flow (Corrected formula)
free_cash_flow = net_income + depreciation_amortization - capex - change

# Present in tabular format
data = {'Item': ['Net Income', 'Depreciation & Amortization', 'Capital E
        'Value': [net_income, depreciation_amortization, capex, change_i

df = pd.DataFrame(data)
print(df)
```

Force the model to write the Python code (or code in some other language) and show the computations for FCF.

Agentic AI Workflow: Valuing a Company



Steps run largely on their own — and after scenarios are built, the agent can loop back to source more data before projecting.

Using AI to find the Peers

- Screening: AI scans filings and business descriptions to surface candidate peers
  - NLP compares product mix, geography and customers, going beyond static SIC/GICS codes
- Clustering: machine learning groups companies by growth, margin and risk profile
  - Flags peers with similar economic drivers even across different industry labels
- Continuous refresh: the peer set updates automatically as business mix or markets shift
  - Stale comps drop out and newly relevant entrants are flagged for review

AI in Action: Finding Peers for Maruti Suzuki

Prompt used: *“Find the closest listed peers to Maruti Suzuki India for relative valuation — same industry, product segment overlap, and similar growth/margin profile. Return names with forward P/E, market cap, and growth.”*

Company	Segment	Fwd P/E	Market Cap	Est. Growth
Maruti Suzuki India	Mass-market PV	33.8x	₹5.0T	14.8%
Mahindra & Mahindra	SUV / UV, Tractors	29.5x	₹4.2T	11.7%
Hyundai Motor India	Mass-market PV	33.1x	₹1.9T	15.0%
Tata Motors (PV)	Mass-market PV / EV	11.7x	₹1.3T	-8.9%
Bajaj Auto	Two/three-wheelers	30.4x	₹2.5T	12.7%

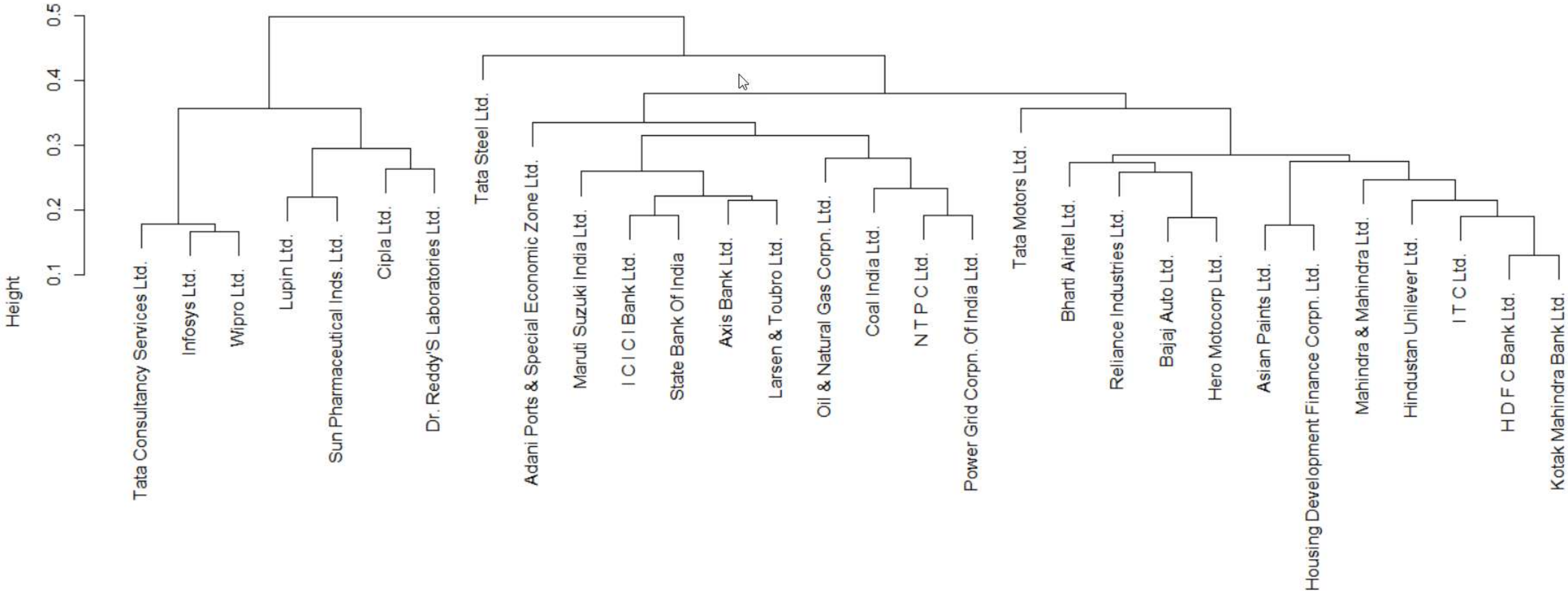
Warren AI (from Investing.com): Finding Peers for Maruti Suzuki

Prompt used: **Find the peer companies for Maruti Suzuki India Limited. And then compare them with Western, their valuation multiples. Prepare a final table to summarize the data.**

Company	Market Cap	P/E (LTM)	EBITDA Margin	Revenue Growth	ROE
Maruti Suzuki India Limited	₹4.30T	25.9x	11.7%	19.9%	14.4%
Mahindra Mahindra Limited	₹3.82T	21.0x	18.6%	25.4%	20.1%
Tata Motors Passenger Vehicles Limited	₹1.19T	30.1x	3.9%	-8.1%	-1.4%
Hyundai Motor India Limited	₹1.78T	25.0x	12.3%	2.3%	29.9%

Company	Market Cap	P/E (LTM)	EBITDA Margin	Revenue Growth	ROE

Company	Market Cap	P/E (LTM)	EBITDA Margin	Revenue Growth	ROE
Bajaj Auto Limited	₹3.19T	27.6x	23.6%	22.7%	29.0%
Hero MotoCorp Limited	₹1.15T	18.6x	14.4%	15.9%	28.1%
TVS Motor Company Limited	₹2.07T	55.4x	14.0%	26.9%	33.4%



Height
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Tata Consultancy Services Ltd.
Infosys Ltd.
Wipro Ltd.
Lupin Ltd.
Sun Pharmaceutical Inds. Ltd.
Cipla Ltd.
Dr. Reddy'S Laboratories Ltd.
Tata Steel Ltd.
Adani Ports & Special Economic Zone Ltd.
Maruti Suzuki India Ltd.
I C I C I Bank Ltd.
State Bank Of India
Axis Bank Ltd.
Larsen & Toubro Ltd.
Oil & Natural Gas Corpn. Ltd.
Coal India Ltd.
N T P C Ltd.
Power Grid Corpn. Of India Ltd.
Tata Motors Ltd.
Bharti Airtel Ltd.
Reliance Industries Ltd.
Bajaj Auto Ltd.
Hero Motocorp Ltd.
Asian Paints Ltd.
Housing Development Finance Corpn. Ltd.
Mahindra & Mahindra Ltd.
Hindustan Unilever Ltd.
I T C Ltd.
H D F C Bank Ltd.
Kotak Mahindra Bank Ltd.

EPS can include one-off gains: **Example: Alphabet's EPS**

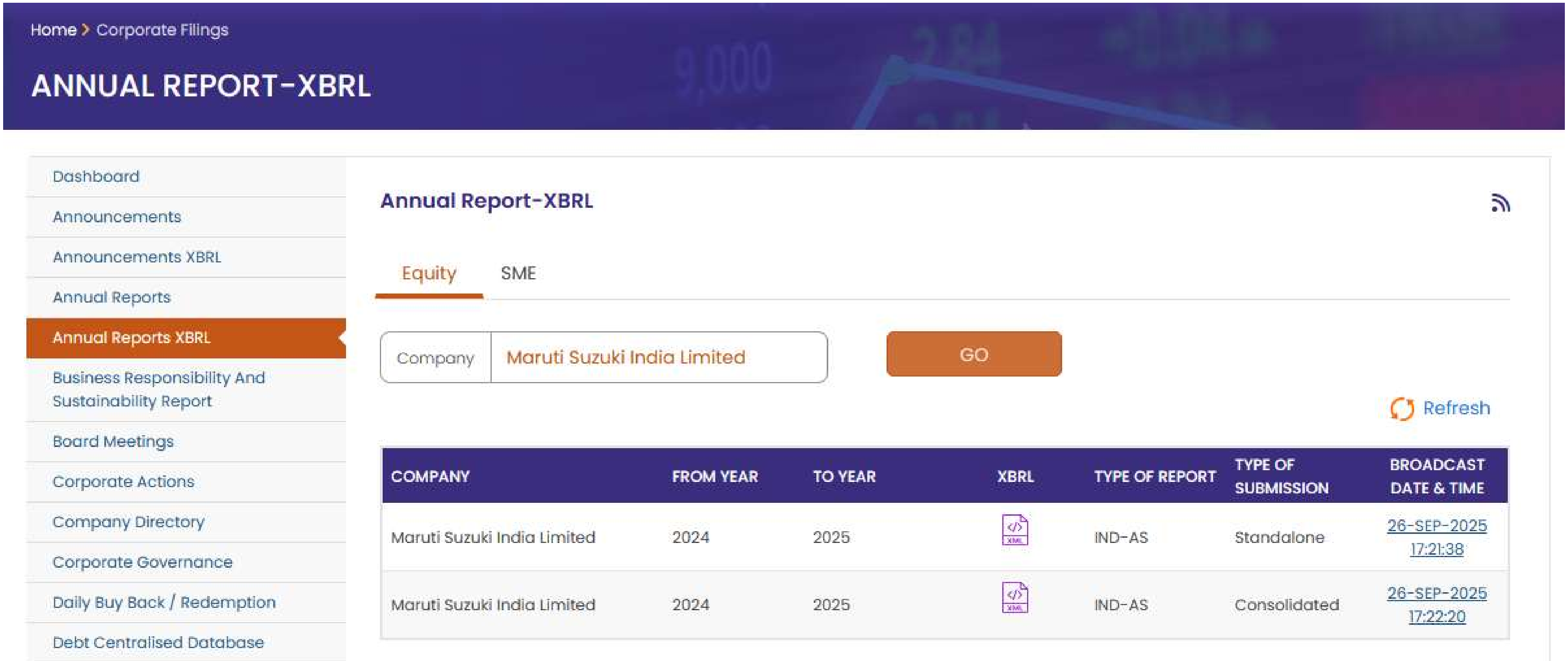
- Q2 2026 diluted EPS: $9.11 (GAAP)
- $99B: unrealized gains, SpaceX + Anthropic stakes
- Adjusted EPS: $2.85 — missed $2.89 consensus

Example: Anthropic's multiple

- Valuation: $965B (Series H, May 2026)
- Q1 annualized: $19.2B → ~50x
- H1 annualized: $31B → ~31x
- Run-rate: $47B → ~20x
    - *Run rate: annualizes the most recent period’s revenue (e.g., latest month × 12) to project a full-year revenue — not actual trailing revenue*

Sourcing Company Data From the Internet

- Stock exchange & regulatory filings
  - NSE, BSE company disclosures; SEBI and MCA filings — financials are filed in XBRL, a structured, tagged format
- Financial data aggregators built for Indian equities
  - Screener.in, Moneycontrol, Trendlyne, ACE Equity, Capitaline
- Company-published sources
  - Investor relations pages: annual reports, quarterly results, investor presentations
- Institutional-grade data providers
  - Bloomberg, Refinitiv, CMIE Prowess — for cross-checking and longer history



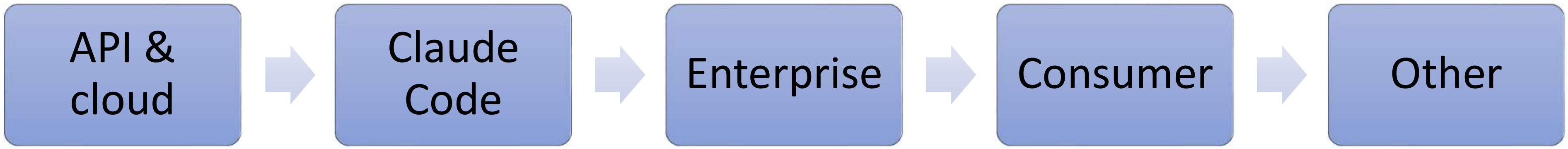
Use any XBRL-to-Excel converter:

1. https://products.aspose.app/finance/conversion/xbrl-to-xlsx
2. Alternatively, drag the XBRL file to Excel, and read it as XML Table while directing reading from Excel

Forecasting Revenue: Don't just grow last year's number

- Naive method: last year's revenue × (1 + growth rate)
- Hides what's driving the number
    - **Build the driver, don't guess the output**
- Break revenue into real components first
- Project each component on its own driver

Example: how Anthropic makes money



Example: Anthropic's revenue break-up

- Revenue in 2026: $42,560M total
- API & cloud 72.8%, Claude Code 11.3%, enterprise 9.3%, consumer 4.2%, other 2.3%

Base-case 2026 revenue mix		
Revenue stream	**Revenue**	**Mix**
API & cloud consumption	$31,000	72.8%
Claude Code	$4,800	11.3%
Enterprise subscriptions	$3,960	9.3%
Consumer subscriptions	$1,800	4.2%
Other partnerships	$1,000	2.3%

Questions to ask while projecting revenue

- Model pricing, efficiency gains
- Competition: OpenAI, Google Gemini, Chinese Models (many are open-weights)
- DoD dispute: several billion in 2026 revenue at risk
  - Pentagon threatened to classify Anthropic as a “supply-chain risk,” jeopardizing its $200 million defense contract and forcing all contractors to cut ties.

Example: Anthropic's Revenue

- Billed token volume × realized blended price

Revenue stream	Revenue	Mix
API & cloud consumption	$31,000	72.8%
Claude Code	$4,800	11.3%
Enterprise subscriptions	$3,960	9.3%
Consumer subscriptions	$1,800	4.2%
Other partnerships	$1,000	2.3%
Total revenue	**$42,560**	**100.0%**

Driver / revenue stream	2026E
API & cloud consumption	
Billed token-equivalent volume (trillion tokens)	3,100.0
Token-volume growth	
Realized blended price ($/MTok)	$10.00
Realized-price change	
API & cloud revenue ($M)	$31,000

Consumer subscriptions	
Paid subscribers (millions)	5.0
Subscriber growth	
Monthly ARPU ($)	$30.00
ARPU growth	
Consumer subscription revenue ($M)	$1,800

Claude Code	
Paid seat equivalents (millions)	3.2
Seat growth	
Monthly ARPU ($)	$125.00
ARPU growth	
Claude Code revenue ($M)	$4,800

Enterprise subscriptions — non-usage	
Paid seats (millions)	6.0
Seat growth	
Monthly ARPU ($)	$55.00
ARPU growth	
Enterprise subscription revenue ($M)	$3,960

Build Scenarios

- Not a label like “optimistic” or “pessimistic”
- A specific set of driver values
- Each driver: a different value per scenario

Three Scenarios for Anthropic's Valuation

Prompt used: *"Model Anthropic's valuation using a DCF approach. Provide three scenarios — bear, base, bull — with distinct assumptions for revenue growth rate, margin trajectory, discount rate (WACC), and terminal growth rate."*

Driver	Bear Case	Base Case	Bull Case
Revenue growth (3-yr CAGR)	~30% — growth decelerates sharply	~45% — growth cools but stays strong	~65% — enterprise & coding demand compounds
Margin trajectory	Stays negative on elevated compute costs	Narrows toward breakeven by ~2028	Turns solidly positive as scale economics kick in
Discount rate (WACC)	~14% — higher risk premium	~11% — in line with high-growth tech peers	~9% — lower risk premium as leadership solidifies
Terminal growth rate	~2.5% — converges to GDP-like growth	~3.5% — durable AI infrastructure demand	~4.5% — AI becomes a larger share of IT spend

Anthropic: Risk Scenarios

Driver	Bear Case	Base Case	Bull Case
DoD case outcome	Designation upheld — loses federal/defense channel	Case narrows — core commercial business unaffected	Anthropic wins — designation lifted
GPU availability	Compute stays scarce, squeezes training budget	Supply eases via custom silicon (TPU, Trainium)	Ample, diversified compute via multi-cloud deals
Chinese competition	Frontier open models undercut pricing globally	Chinese models compete on cost; Anthropic keeps enterprise edge	Western enterprises stay locked out on security grounds
Revenue (FY2027E)	**~$90–110B**	**~$150–180B**	**~$220–260B**
Operating margin (FY2027E)	**-15% to -25%**	**-5% to 0%**	**+10% to +15%**

Example: Anthropic's scenarios

- Bear: 2,600 trillion tokens, $8.75/MTok
- Base: 3,100 trillion tokens, $10.00/MTok
- Bull: 3,600 trillion tokens, $10.50/MTok

Anthropic Revenue Model — Scenario Inputs

Blue-font cells are editable assumptions. Yellow cells are the most uncertain 2026 starting points. All monetary outputs are USD millions unless noted.

Revenue stream	Driver / assumption	Bear	Base	Bull	Basis	Source / modeling note
API & cloud consumption	2026 billed token-equivalent volume (trillion tokens)	2,600.0	3,100.0	3,600.0	Analyst assumption	Absolute company-wide token volume is not disclosed; calibrated against observed revenue anchors.
API & cloud consumption	2026 realized blended price ($ per MTok)	$8.75	$10.00	$10.50	Analyst assumption	List pricing varies by model, input/output mix, caching and cloud channel. This is a realized blended rate.
API & cloud consumption	2027 token-volume growth	35.0%	50.0%	70.0%	Analyst assumption	Volume growth is separated from price/mix change.
API & cloud consumption	Annual change in token-volume growth	(7.0%)	(9.0%)	(11.0%)	Analyst assumption	Percentage-point annual deceleration.
API & cloud consumption	Minimum token-volume growth	8.0%	10.0%	15.0%	Analyst assumption	Long-run floor through the explicit forecast period.
API & cloud consumption	Annual realized-price change	(8.0%)	(5.0%)	(3.0%)	Analyst assumption	Captures model price compression, mix and efficiency.
Claude Code	2026 paid seat equivalents (millions)	2.2	3.2	4.2	Analyst assumption	Calibrated to the disclosed >$2.5B annualized revenue in Feb 2026 and subsequent growth.
Claude Code	2026 monthly ARPU ($)	$95.00	$125.00	$145.00	Analyst assumption	Blended individual and enterprise-equivalent revenue per paid seat.
Claude Code	2027 seat growth	35.0%	55.0%	80.0%	Analyst assumption	Business subscriptions reportedly quadrupled early in 2026; forecast assumes rapid deceleration.
Claude Code	Annual change in seat growth	(7.0%)	(10.0%)	(13.0%)	Analyst assumption	Percentage-point annual deceleration.
Claude Code	Minimum seat growth	8.0%	10.0%	15.0%	Analyst assumption	Long-run floor through 2031.
Claude Code	Annual ARPU growth	-	2.0%	4.0%	Analyst assumption	Captures enterprise mix and product depth.

Forecasting Expenses

Cost area	Driver / assumption	Bear	Base	Bull	Modeling note
Inference / delivery	2026 cash inference cost ($ per MTok)	$7.00	$6.50	$6.00	Includes cloud capacity, accelerators, electricity and networking used to serve API demand.
Inference / delivery	Annual inference cost-per-token change	(12.0%)	(18.0%)	(24.0%)	Efficiency improves through hardware, software, batching, caching and utilization.
Inference / delivery	Non-API delivery cost as % non-API revenue	15.0%	12.0%	10.0%	Covers product hosting, support and distribution for Claude Code and subscriptions.
Inference / delivery	Support and safety delivery cost as % revenue	3.0%	2.5%	2.0%	Recurring reliability, monitoring and customer-serving safety costs.

Example: Maruti's business

- Volume mix: small car, SUV, premium
- SUV +44.6%, premium +28%
- Royalty to Suzuki Motor Corp: 5–6% of net sales

Example: Maruti's Revenue

- Domestic volume × blended realization

Base Scenario					
Driver / revenue stream	**FY27E**	**FY28E**	**FY29E**	**FY30E**	**FY31E**
Domestic					
Volume (thousand units)	2,136.0	2,328.2	2,507.5	2,668.0	2,804.1
Volume growth		9.0%	7.7%	6.4%	5.1%
Realization (₹ lakh/unit)	8.52	8.78	9.04	9.31	9.59
Realization growth		3.0%	3.0%	3.0%	3.0%
Domestic revenue (₹cr)	181,987	204,317	226,651	248,391	268,891

Example: Maruti's scenarios

Maruti Suzuki Revenue Model — Scenario Inputs

Blue-font cells are editable assumptions. Yellow cells are the most uncertain FY27 starting points. Volumes in thousand units, realization in ₹ lakh/unit unless noted.

Revenue stream	Driver / assumption	Bear	Base	Bull	Basis	Source / modeling note
Domestic	FY27E volume (thousand units, annualized)	1,950	2,136	2,260	Analyst assumption (annualized)	Q1 FY27 domestic wholesale >534,000 units × 4; point-in-time annualization, not a confirmed full-year actual.
Domestic	FY28E volume growth	5.0%	9.0%	14.0%	Analyst assumption	Q1 FY27 domestic growth was 33% YoY off an easy prior-year base; assumes normalization.
Domestic	Annual change in volume growth	-1.0%	-1.3%	-1.6%	Analyst assumption	Percentage-point annual deceleration.
Domestic	Minimum volume growth	3.0%	5.0%	7.0%	Analyst assumption	Long-run floor through FY31.
Domestic	FY27E blended realization (₹ lakh/unit)	8.10	8.52	8.90	Derived from disclosed revenue	Consolidated Q1 FY27 revenue (₹ 52,469.8cr) ÷ estimated Q1 total volume (~616,000 units — see Source Data).
Domestic	Annual realization growth	1.0%	3.0%	5.0%	Analyst assumption	Mix shift toward SUV/premium (SUV +44.6%, premium +28% in Q1 FY27) and pricing power.
Exports	FY27E volume (thousand units, estimated)	240	290	335	Analyst assumption (estimated)	Export units not separately disclosed by Maruti. Anchored to FY24's export share (~13.3% of total volume), grown at the disclosed +28.6% YoY rate.
Exports	FY28E volume growth	8.0%	14.0%	20.0%	Analyst assumption	Global demand, new export models (e.g. flex-fuel variants).
Exports	Annual change in volume growth	-1.5%	-2.0%	-2.5%	Analyst assumption	Percentage-point annual deceleration.
Exports	Minimum volume growth	3.0%	5.0%	8.0%	Analyst assumption	Long-run floor through FY31.
Exports	Annual realization growth	1.0%	3.0%	5.0%	Analyst assumption	Maruti does not disclose a domestic/export price differential; same per-unit trend applied as domestic.
Memo (not in revenue build)	Royalty to Suzuki Motor Corp, % of net sales	5.0%	5.5%	6.0%	Historical disclosed range	Rupee-denominated since 2023. A deduction affecting minority-shareholder value, not a reduction to reported revenue — shown for governance context only.

Audit before you submit

- A finished-looking valuation can still be wrong
- Check every number against its source
- Find the error, explain it, fix it